

Timer



14 Oct 2025

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1. Timer Modules Overview

The STM32 supports following timer modules:

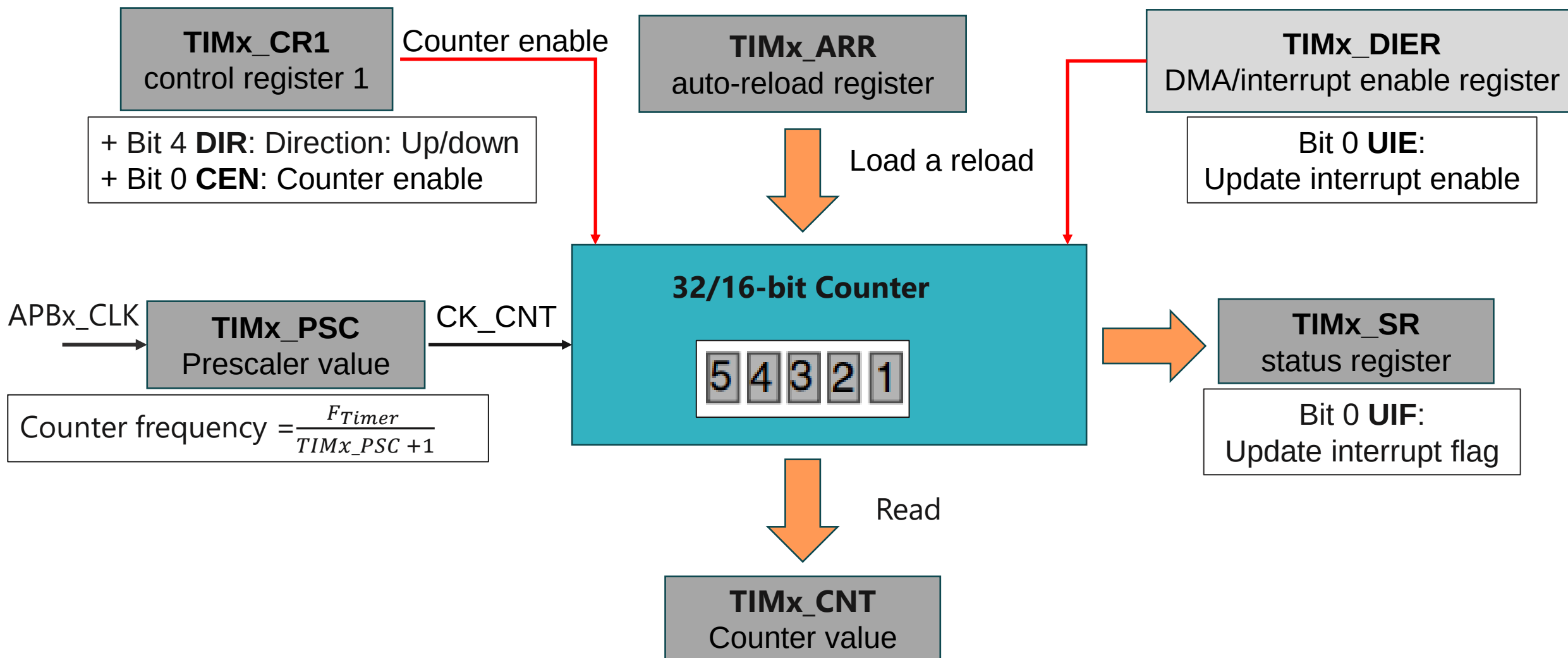
- + Advanced-control timers (TIM1 and TIM8)
- + General-purpose timers (TIM2 to TIM5)
- + General-purpose timers (TIM9 to TIM14)
- + Basic timers (TIM6 and TIM7)
- + SysTick timer(Core Arm M)

1. Timer Modules Overview

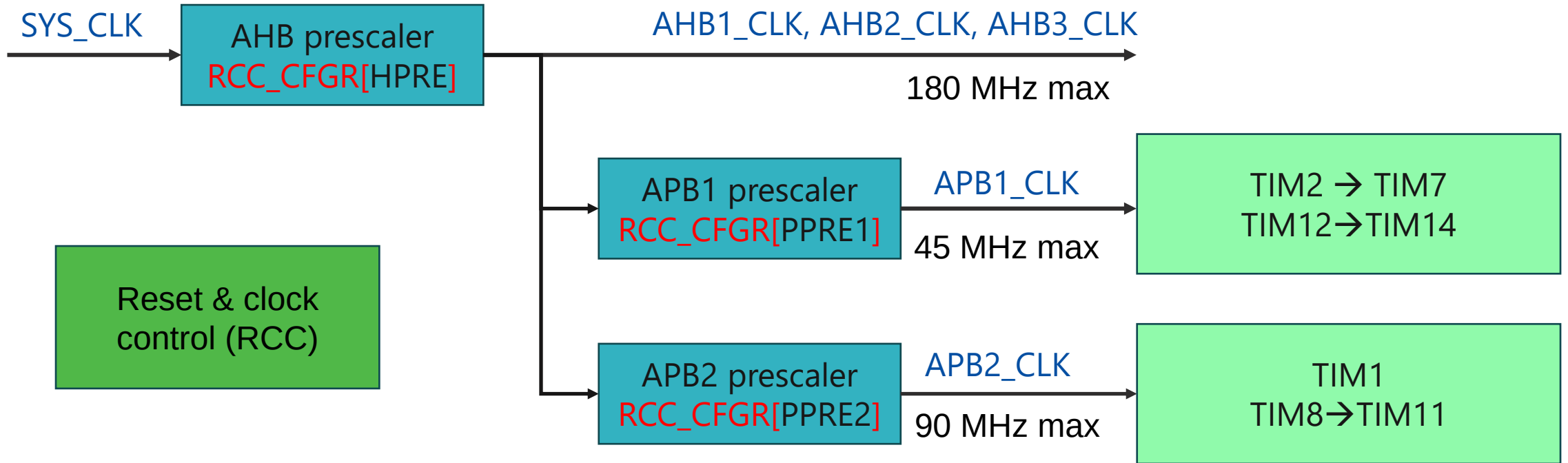
Timer in STM32 is a hardware counter used for:

- Measuring time (creating delays)
- Generating periodic interrupts
- PWM generation
- Input capture(Measuring pulse width)
- Output compare
- Working with encoders (reading rotational position)
- One-pulse mode output...

2. Timer Programming Model



3.1. Setting Clock



3.2. Auto-reload value

17.4.12 TIM1 and TIM8 auto-reload register (TIMx_ARR)

Address offset: 0x2C

Reset value: 0xFFFF

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ARR[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

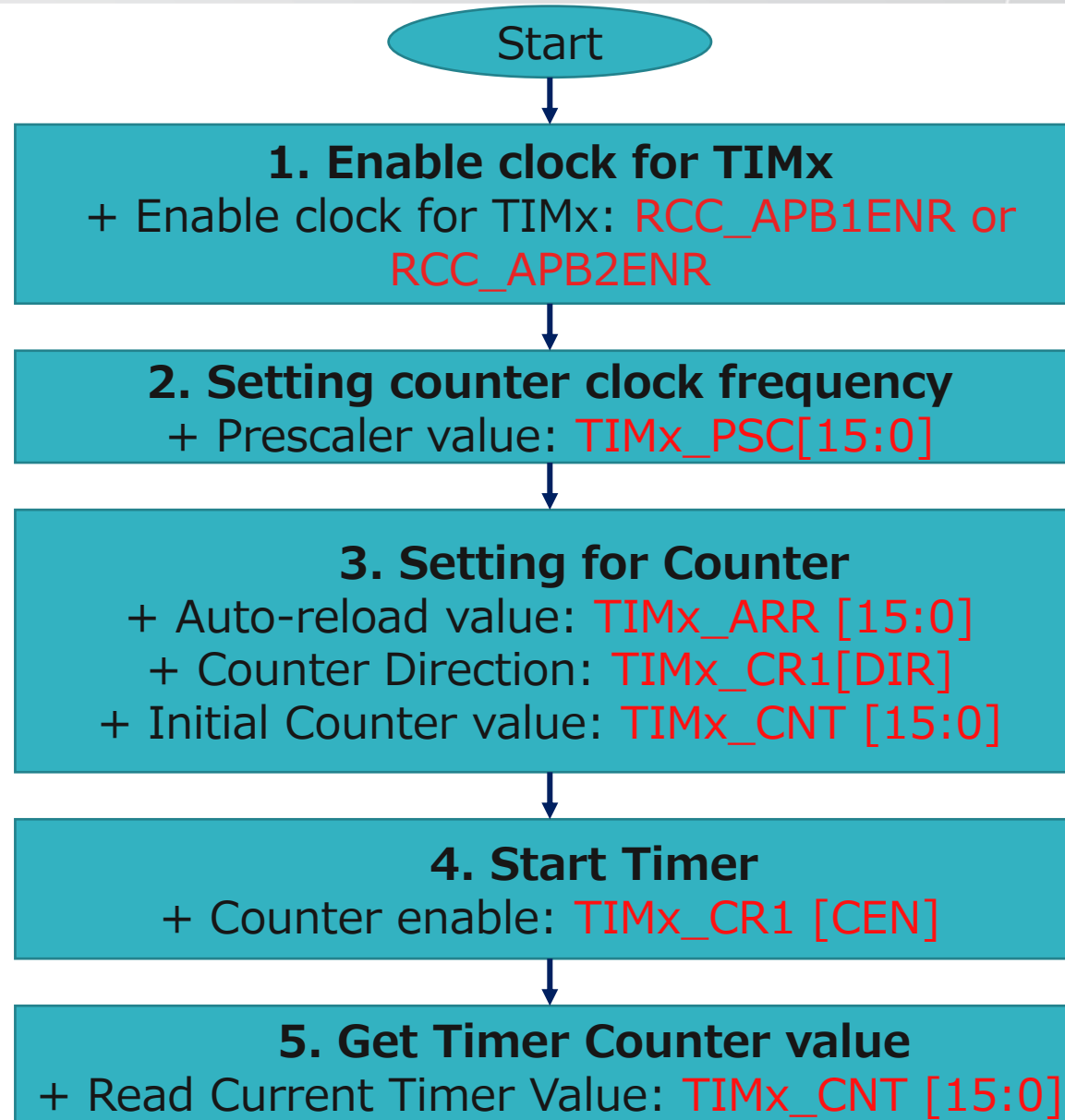
Bits 15:0 **ARR[15:0]**: Auto-reload value

ARR is the value to be loaded in the actual auto-reload register.

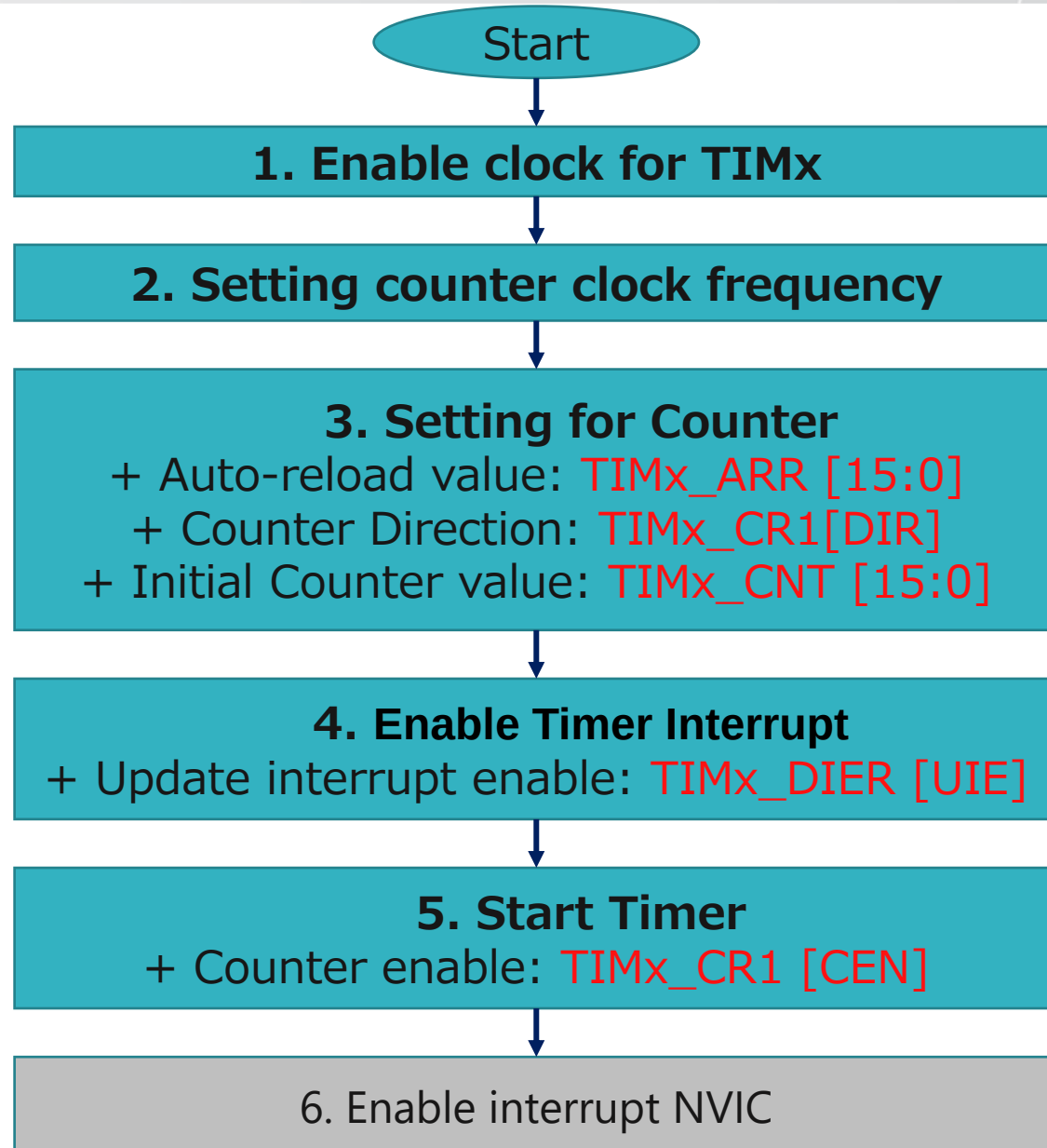
$$\text{Auto-reload value} = \text{Interval(s)} * F_{\text{Counter}}(\text{Hz}) - 1$$

- Calculate the ARR value so that the counting time from 0 to ARR equals 500 ms.
 - + Counter used as upcounter
 - + $F_{\text{Timer}} = 8\text{Mhz}$
 - + Prescaler value = 199

3.4. Configuration for TIMx



3.5. Configuration for interrupt of TIMx



Performance Measurement

Measure the CPU time to complete this for statement.

```
for( unsigned int i = 0; i < 0x3FFFFFF; i++ );
```

4.2. Assignment 2

Blink Led every 500ms using polling technique

4.3. Assignment 3

Blink Led every 500ms using interrupt technique

3.4. Assignment 4

Write a program to simulate a watcher.
hh:mm:ss

3.5. Assignment 5

Write the delay function :
+ void delay(unsigned int ms);
+ ms: millisecond

A nighttime cityscape featuring a prominent skyscraper with a spire, illuminated by warm lights. The building is framed by a large, semi-transparent, stylized letter 'R' in a light purple/pink hue. Other buildings and a body of water are visible in the background under a dark sky with some clouds.

Thank you